Examiner only

4. (a) Radioactive waste contains a mixture of isotopes with a variety of half-lives.
The isotopes emit alpha, beta and gamma radiation.
Discuss the safe storage of radioactive waste.
Include in your answer:

- a description of alpha, beta and gamma radiation,
- the properties of alpha, beta and gamma radiation,
- how radioactive waste could be safely stored. [6 QER]

(b) One isotope of uranium (U) decays naturally by alpha (α) decay into thorium (Th) which in turn decays into protactinium (Pa) by beta (β) decay.

Complete the decay equation for this decay. [4]

$$\overset{\cdots\cdots}{\cdots\cdots}\mathrm{U} \rightarrow {}^{231}_{90}\mathrm{Th} + \overset{4}{\cdots\cdots}\alpha$$

$${}^{231}_{90}\mathrm{Th} \rightarrow \overset{\cdots\cdots}{\cdots\cdots}\mathrm{Pa} + \overset{\cdots\cdots}{-1}\beta$$

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